Team Name:

Experiment Name: Hospital Management System

Objective:

The main objective of this project is to develop a web-based Hospital Management System to simplify and manage hospital activities efficiently.

Specific objectives include:

1. To manage patient and doctor information.
2. To handle patient registration and login.
3. To manage doctor appointments and schedules.
4. To reduce manual paperwork and errors.
5. To provide a user-friendly interface for patients, doctors, and administrators.
6. To ensure secure management of hospital records.

Requirement:

Functional Requirements:

1. Patient registration and login.
2. Patient profile management.
3. Doctor information management.
4. Appointment booking and management.
5. Admin dashboard for managing hospital activities.
6. Contact and inquiry management.

Non-Functional Requirements:

1. User-friendly interface.
2. Secure data management.
3. Fast response time.
4. Reliable database operations.
5. Responsive design for different devices.

Design:

The Hospital Management System follows Django's MVT (Model-View-Template) architecture.

The system consists of three main components:

1. Model: Manages database operations using SQLite.
2. View: Handles application logic and user requests using Django.
3. Template: Displays the user interface using HTML, CSS, Bootstrap, and JavaScript.

Main System Modules:

- Patient Management
- Doctor Management
- Appointment Management
- Admin Dashboard
- User Authentication
- Contact Management

Source Code:

1. **index.html:**

```
{% extends "navigationbar.html" %}
{% load static %}
{% block body %}

<section class="hero">
    <div class="container">
        <div class="row">
            <div class="col-lg-7">

                <h1>
                    Your Health Is Our
```

```
                    <span style="color:#38bdf8;">Priority</span>
                </h1>

                <p>
                    Providing quality healthcare services with
                    experienced doctors, modern technology and
                    compassionate care. Your health and happiness
                    are our greatest responsibility.
                </p>

                <a href="{% url 'login' %}"
                   class="btn btn-info btn-lg text-white me-2">
                    Book Appointment
                </a>

                <a href="#services"
                   class="btn btn-outline-light btn-lg">
                    Explore Services
                </a>

            </div>
        </div>
    </div>
</section>


<!-- ABOUT SECTION -->

<section class="about-section">
    <div class="container">
        <div class="row align-items-center">

            <div class="col-lg-6 mb-4">
                <img src="{% static 'images/hospital1.webp' %}"
                     class="img-fluid rounded shadow"
                     alt="Our Hospital">
            </div>

            <div class="col-lg-6">

                <h5 class="text-info">ABOUT OUR HOSPITAL</h5>
```

```
                <h2 class="fw-bold mb-4">
                    We Are Here To Care For Your Health
                </h2>

                <p class="text-muted">
                    Our hospital is committed to delivering
                    comprehensive healthcare services to patients.
                    We combine medical expertise, advanced
                    technology and compassionate care to provide
                    the best possible treatment.
                </p>

                <p class="text-muted">
                    Our dedicated team of doctors, nurses and
                    healthcare professionals works together
                    to ensure every patient receives personalized
                    attention and quality medical care.
                </p>

                <a href="#services"
                   class="btn btn-primary mt-3">
                    Learn More
                </a>

            </div>
        </div>
    </div>
</section>


<!-- SERVICES SECTION -->

<section class="services-section" id="services">
    <div class="container">

        <div class="section-title2">
            <h2>Our Medical Services</h2>
            <p>
                We provide comprehensive healthcare services
                for you and your family.
            </p>
        </div>
```

```
<div class="row g-4">

    <div class="col-md-4">
        <div class="service-card">
            <div class="service-icon">
                <i class="fas fa-heartbeat"></i>
            </div>
            <h4>Cardiology</h4>
            <p class="text-muted">
                Specialized heart care and cardiovascular
                treatment from experienced professionals.
            </p>
        </div>
    </div>

    <div class="col-md-4">
        <div class="service-card">
            <div class="service-icon">
                <i class="fas fa-user-md"></i>
            </div>
            <h4>Expert Doctors</h4>
            <p class="text-muted">
                Consult qualified and experienced doctors
                for professional medical advice.
            </p>
        </div>
    </div>

    <div class="col-md-4">
        <div class="service-card">
            <div class="service-icon">
                <i class="fas fa-ambulance"></i>
            </div>
            <h4>Emergency Care</h4>
            <p class="text-muted">
                Emergency medical assistance and
                dedicated patient support.
            </p>
        </div>
    </div>

    <div class="col-md-4">
```

```
                <div class="service-card">
                    <div class="service-icon">
                        <i class="fas fa-x-ray"></i>
                    </div>
                    <h4>Diagnostic Services</h4>
                    <p class="text-muted">
                        Modern diagnostic facilities for
                        accurate medical examinations.
                    </p>
                </div>
            </div>

            <div class="col-md-4">
                <div class="service-card">
                    <div class="service-icon">
                        <i class="fas fa-tooth"></i>
                    </div>
                    <h4>Dental Care</h4>
                    <p class="text-muted">
                        Complete dental care and oral health
                        services for patients.
                    </p>
                </div>
            </div>

            <div class="col-md-4">
                <div class="service-card">
                    <div class="service-icon">
                        <i class="fas fa-baby"></i>
                    </div>
                    <h4>Child Care</h4>
                    <p class="text-muted">
                        Specialized healthcare and medical
                        support for children.
                    </p>
                </div>
            </div>

        </div>
    </div>
</section>

<!-- DOCTORS SECTION -->
```

```
<section class="services-section bg-light" id="doctors">
    <div class="container">

        <div class="section-title">
            <h2>Meet Our Doctors</h2>
            <p class="text-muted">
                Our experienced medical professionals
                are dedicated to your health.
            </p>
        </div>

        <div class="row g-4">

            <div class="col-md-4">
                <div class="doctor-card">
                    <img src="{% static 'images/doctor1.jpg' %}"
                         alt="Doctor">
                    <div class="card-body">
                        <h5>Dr. Bruce Banner</h5>
                        <p class="text-muted">
                            Cardiologist
                        </p>
                    </div>
                </div>
            </div>

            <div class="col-md-4">
                <div class="doctor-card">
                    <img src="{% static 'images/doctor2.jpg' %}"
                         alt="Doctor">
                    <div class="card-body">
                        <h5>Dr. Victor von Doom</h5>
                        <p class="text-muted">
                            Neurologist
                        </p>
                    </div>
                </div>
            </div>

            <div class="col-md-4">
                <div class="doctor-card">
                    <img src="{% static 'images/doctor3.avif' %}"
```

```
                        alt="Doctor">
                    <div class="card-body">
                        <h5>Dr. Stephen Strange</h5>
                        <p class="text-muted">
                            Pediatrician
                        </p>
                    </div>
                </div>
            </div>

        </div>
    </div>
</section>
<!-- FOOTER -->

<footer>
    <div class="container">

        <div class="row">

            <div class="col-md-4 mb-4">
                <h4>HealthCare Hospital</h4>
                <p>
                    Your trusted healthcare partner.
                    Providing quality medical services
                    with care and compassion.
                </p>
            </div>

            <div class="col-md-4 mb-4">
                <h5>Quick Links</h5>
                <p><a href="#">Home</a></p>
                <p><a href="#services">Our Services</a></p>
                <p><a href="#doctors">Doctors</a></p>
                <p><a href="{% url 'login' %}">Login</a></p>
            </div>

            <div class="col-md-4 mb-4">
                <h5>Contact Us</h5>
                <p>Address: Kallyanpur, Dhaka</p>
                <p>Phone: +880 1877-804699</p>
                <p>Email: info@healthcare.com</p>
            </div>
```

```
        </div>

        <hr>

        <p class="text-center mb-0">
            &copy; 2026 HealthCare Hospital.
            All Rights Reserved.
        </p>

    </div>
</footer>



{% endblock %}
```

2. **Navigationbar.html**

```
<html>
    <head>
        {% load static %}
        <title>Hospital Management System</title>
        <meta charset="utf-8">
        <meta name="viewport" content="width=device-width, initial-scale=1">
        <link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/4.3.1/css/bootstrap.min.css">
        <link rel="stylesheet"
href="https://use.fontawesome.com/releases/v5.6.3/css/all.css" integrity="sha384-
UHRtZLI+pbxtHCWp1t77Bi1L4ZtiqrqD80Kn4Z8NTSRyMA2Fd33n5dQ8lWUE00s/"
crossorigin="anonymous"></head>
        <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.5.2/css/all.min.css">
        <script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script>
        <script
src="https://cdnjs.cloudflare.com/ajax/libs/popper.js/1.16.0/umd/popper.min.js"><
/script>
```

```
        <script
src="https://maxcdn.bootstrapcdn.com/bootstrap/4.3.1/js/bootstrap.min.js"></scrip
t>
        <link rel="stylesheet" type="text/css" href="{% static
'css/mystyle.css' %}">

        <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
        <link rel="stylesheet"
href="https://cdn.datatables.net/1.13.8/css/jquery.dataTables.min.css">
        <link rel="stylesheet"
href="https://cdn.datatables.net/buttons/2.4.2/css/buttons.dataTables.min.css">
        <script
src="https://cdn.datatables.net/1.13.8/js/jquery.dataTables.min.js"></script>
        <script
src="https://cdn.datatables.net/buttons/2.4.2/js/dataTables.buttons.min.js"></scr
ipt>
        <script
src="https://cdn.datatables.net/buttons/2.4.2/js/buttons.html5.min.js"></script>
        <script
src="https://cdnjs.cloudflare.com/ajax/libs/jszip/3.10.1/jszip.min.js"></script>
        <script
src="https://cdnjs.cloudflare.com/ajax/libs/pdfmake/0.2.7/pdfmake.min.js"></scrip
t>
        <script
src="https://cdnjs.cloudflare.com/ajax/libs/pdfmake/0.2.7/vfs_fonts.js"></script>

        <script>
            $(document).ready(function () {
                $('#example').DataTable({
                    dom: 'Bfrtip',
                    buttons: [
                        'copyHtml5',
                        'excelHtml5',
                        'csvHtml5',
                        'pdfHtml5'
                    ]
                });
            });
        </script>


    </head>
    <body>
```

```
        <nav class="navbar navbar-expand-sm bg-dark navbar-dark">
            <div class="container">
                <a class="navbar-brand" href="{% url 'home' %}">HealthCare
Hospital</a>
                <button class="navbar-toggler" type="button" data-
toggle="collapse" data-target="#myMenu">
                    <span class="navbar-toggler-icon"></span>
                </button>
                <div class="collapse navbar-collapse" id="myMenu">
                    <ul class="navbar-nav text-center ml-auto">
                        {% if not request.user.is_staff %}
                            <li class="nav-item"><a class = "nav-link" href="{%
url 'home' %}"><i class="fas fa-home"></i> Home</a></li>
                            <li class="nav-item"><a class = "nav-link" href="{%
url 'about' %}"><i class="fas fa-info-circle"></i> About</a></li>
                            <li class="nav-item"><a class = "nav-link" href="{%
url 'contact' %}"><i class="fas fa-address-card"></i> Contact</a></li>
                            <li class="nav-item"><a class = "nav-link" href="{%
url 'login' %}"><i class="fas fa-sign-in-alt"></i> Login</a></li>
                        {% endif %}

                        {% if request.user.is_staff %}
                            <li class="nav-item"><a class = "nav-link" href="{%
url 'admin_dashboard' %}"><i class="fas fa-home"></i> Home</a></li>
                            <li class="nav-item dropdown"><a class = "nav-link
dropdown-toggle" data-toggle="dropdown" href="#"><i class="fas fa-user-md"></i>
Doctor</a>
                                <div class="dropdown-menu">
                                    <a class = "dropdown-item" href="{% url
'add_doctor' %}">Add Doctor</a>
                                    <a class = "dropdown-item" href="{% url
'view_doctor' %}">View Doctor</a>
                                </div>
                            </li>

                            <li class="nav-item dropdown"><a class = "nav-link
dropdown-toggle" data-toggle="dropdown" href="#"><i class="fas fa-bed"></i>
Patient</a>
                                <div class="dropdown-menu">
                                    <a class = "dropdown-item" href="{% url
'add_patient' %}">Add Patient</a>
                                    <a class = "dropdown-item" href="{% url
'view_patient' %}">View Patient</a>
                                </div>
```

```
                            </li>

                            <li class="nav-item dropdown"><a class = "nav-link
dropdown-toggle" data-toggle="dropdown" href="#"><i class="fas fa-book-
medical"></i> Appointment</a>
                                <div class="dropdown-menu">
                                    <a class = "dropdown-item" href="{% url
'add_appointment' %}">Add Appointment</a>
                                    <a class = "dropdown-item" href="{% url
'view_appointment' %}">View Appointment</a>
                                </div>
                            </li>

                            <li class="nav-item"><a class = "nav-link" href="{%
url 'logout' %}"><i class="fas fa-sign-out-alt"></i> Logout</a></li>
                        {% endif %}

                    </ul>
                </div>
            </div>
        </nav>
        {% block body %}

        {% endblock %}

    </body>
</html>
```

System Snapshots: